

A Kerr microring resonator as the policy of a reinforcement-learning agent

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I. KEY HYPOTHESIS

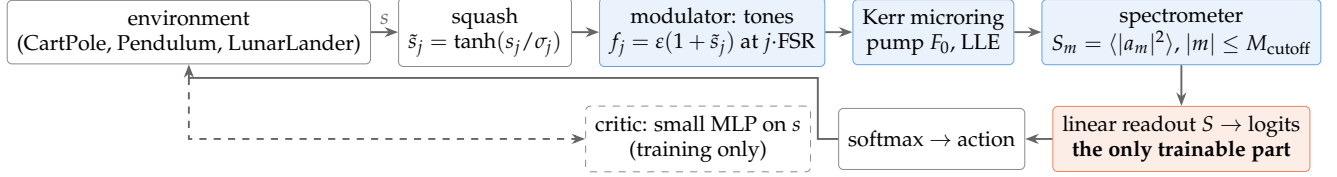


FIG. 1: The optical policy. Blue boxes are optics (nothing in them is trained); the orange box holds the 36 (CartPole), 54 (Pendulum) or 136 (LunarLander) trainable weights. PPO trains the readout and the critic; the ring is simulated by the Lugiato–Lefever equation.

Microring optical resonators are intrinsically noisy, and their algorithmic utility remains unclear. Although such systems can in principle be scaled to large photonic networks, noise, together with efficient schemes for encoding and decoding information, remains a primary challenge. To this end, in our earlier exploratory numerical study of optical reservoir computing with chaotic frequency combs (code repository: <https://github.com/PashaDolgirev/rc-chaotic-comb>), a trivial last-symbol predictor outperformed the comb-based computation, underscoring the need to identify tasks for which the optical dynamics provide genuine computational leverage.

Here, we asked whether an optical microring resonator can serve as the *policy* of a reinforcement-learning agent: the observation of the environment is written into the light that drives the ring, the ring’s output spectrum is read by photodetectors, and the only trainable element is a single linear layer of a few dozen weights that turns detected line powers into an action – see Fig. 1 for the full pipeline. We consider three standard, low-dimensional control environments – CartPole, Pendulum, and LunarLander – and show that the optical policy, trained with an unmodified proximal-policy-optimisation (PPO) loop, can outperform a linear-policy baseline. A multilayer perceptron (MLP) achieves better performance, as expected, but our goal here is different: to establish that noisy nonlinear optical dynamics can be trained and controlled sufficiently well to solve nontrivial RL tasks.

The next steps include demonstrating the approach experimentally, extending it to networks of coupled resonators, developing scalable optical encoding and readout schemes, and jointly optimizing trainable hardware parameters. These steps will determine whether nonlinear photonic dynamics can provide a practical computational substrate for more ambitious reinforcement-learning tasks.

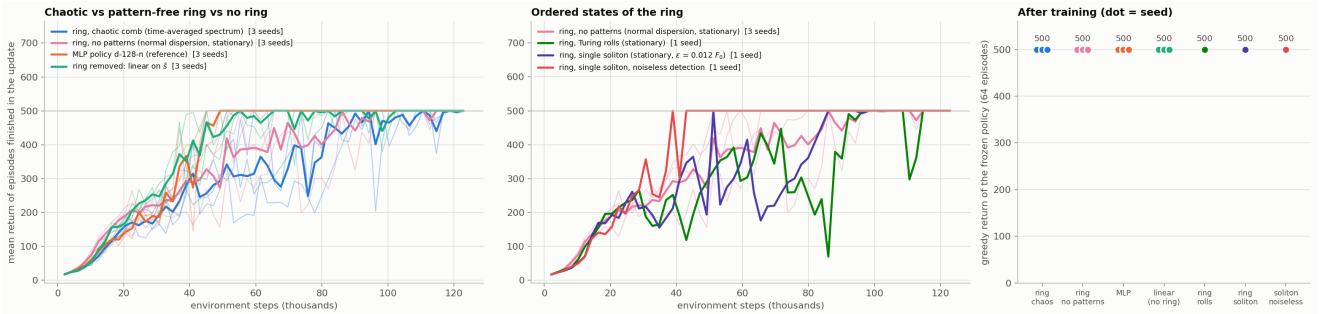


FIG. 2: CartPole-v1. Left: learning curves (mean return of the episodes finished in each update; thin lines are single seeds, bold the seed average) for the chaotic and the pattern-free ring against the same PPO loop with a neural-network policy and with the ring removed (a linear layer applied to the same squashed inputs). Middle: the ordered states. Right: greedy return of the frozen final policies over 64 fresh episodes.

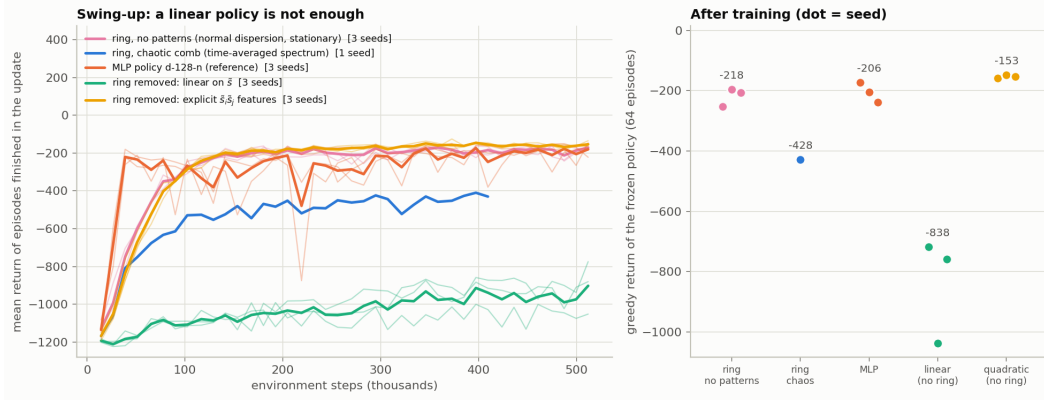


FIG. 3: Results—learning curves and frozen-policy returns—for Pendulum swing-up. The linear policy (green) performs worst, falling below the chaotic-comb policy, because it can only represent a single linear decision boundary in $(\cos \theta, \sin \theta, \dot{\theta})$.

II. INFORMATION ENCODING AND DECODING

We model the optical micro-ring via the dimensionless Lugiato–Lefever equation (LLE):

$$\partial_t \psi = -(1 + i\Delta)\psi + id_2 \partial_\varphi^2 \psi + i|\psi|^2 \psi + F(\varphi), \quad (1)$$

with time in units of the photon lifetime $2/\kappa$, detuning Δ and dispersion d_2 ; $\psi = \sum_m a_m e^{im\varphi}$ and $|a_m|^2$ is the power in comb line m . The simulator is a PyTorch analogue of the JAX solver of the companion repository `rc-chaotic-comb`.

Information Encoding—Following Fig. 1, each component s_j of the observation ($d = 4$ for CartPole, 3 for Pendulum, and 8 for LunarLander) is squashed to $\tilde{s}_j \in [-1, 1]$ and encoded in the amplitude of a *sub-band tone* driving mode $m = j$:

$$F(\varphi) = F_0 + \sum_{j=1}^d \varepsilon (1 + \tilde{s}_j) e^{ij\varphi}, \quad \varepsilon \leq 0.32 F_0.$$

The offset $(1 + \tilde{s}_j)$ is essential. For a drive

$$F(\theta) = F_0 + \sum_j f_j e^{im_j \varphi},$$

the LLE is invariant under rotations of the ring: if $\psi(\varphi, t)$ is a solution for tones $\{f_j\}$, then $\psi(\varphi - \varphi_0, t)$ is a solution for $\{f_j e^{-im_j \varphi_0}\}$, with mode amplitudes $a_m \rightarrow a_m e^{-im \varphi_0}$ and hence unchanged line intensities $|a_m|^2$. Thus, if the sign of $\tilde{s}_j$ carries information, a purely signed encoding can make distinct inputs spectrally indistinguishable. For example, choosing $\varphi_0 = \pi$ gives $f_j \rightarrow (-1)^{m_j} f_j$, so that

$$(\tilde{s}_1, \tilde{s}_2, \tilde{s}_3, \tilde{s}_4) \quad \text{and} \quad (-\tilde{s}_1, \tilde{s}_2, -\tilde{s}_3, \tilde{s}_4) \quad (2)$$

produce exactly identical spectra. For CartPole, with $(x, \dot{x}, \theta, \dot{\theta})$ mapped to $m = 1, 2, 3, 4$, respectively, the ring therefore cannot distinguish “cart left, pole leaning left” from “cart right, pole leaning right” using intensity readout alone. In particular, sign θ is not recoverable from the spectrum. Consistent with this symmetry, PPO with the unshifted encoding stalls near a return of 100, whereas the shifted encoding resolves this issue and reaches the maximal return of 500.

In our simulations, we use squashing scales $\sigma = (1, 0.75, 0.075, 0.75)$ for $(x, \dot{x}, \theta, \dot{\theta})$, $(1, 1, 8)$ for $(\cos \theta, \sin \theta, \dot{\theta})$ and $(0.6, 0.8, 0.8, 0.8, 0.5, 0.6, 1, 1)$ for the LunarLander state. Features are standardised with mean and standard deviation from a task-independent sweep of the inputs over $[-1, 1]^d$ before training; the readout starts at zero (uniform random policy).

Information Decoding—For chaotic combs, we follow the averaging protocol of `rc-chaotic-comb`. After each change of the input tones, the first $T_{\text{relax}} = 3$ lifetimes are discarded, after which the line powers are sampled every

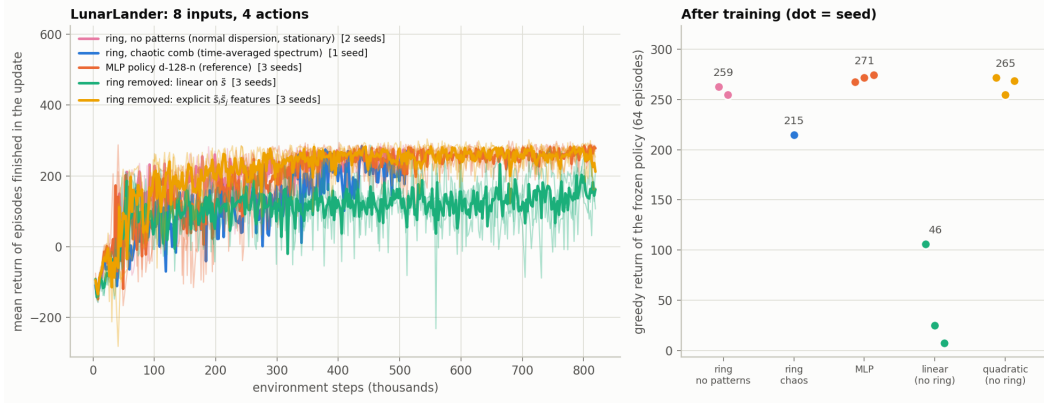


FIG. 4: LunarLander-v3: learning curves (the ring runs were stopped after 512k steps, the baselines after 819k) and frozen returns.

0.05 for $T_{\text{avg}} = 25$ lifetimes. The resulting time-averaged spectrum, $\langle |a_m|^2 \rangle$, forms the feature vector entering the linear readout of Fig. 1. For stationary regimes—the pattern-free state, Turing rolls, and the single-soliton state—no temporal averaging is required; after each change of the tones, the relaxed line powers are used directly as the feature vector.

Our numerical explorations indicate that pattern-forming states (Turing rolls and solitons) are too fragile for robust computation: they tolerate only tones weak enough that the ring behaves essentially as a linear transducer, while beyond this window the branch is lost and training fails completely. The chaotic comb and the pattern-free state remain robust under substantially stronger drive, with the pattern-free state performing slightly better overall. Nevertheless, the fact that a chaotic comb can serve as an RL policy at all is itself an intriguing result.

PPO—64 environments $\times$ 32 steps per update, GAE $\lambda = 0.95$, clip 0.2, 20 policy epochs and 20 critic iterations per buffer, Adam 10^{-2} ; $\gamma = 0.99$ and 60 updates (123k steps) for CartPole, $\gamma = 0.95$, reward scaled by 0.1 and 250 updates (512k steps) for the pendulum (200 for the chaotic ring); $\gamma = 0.99$, reward scaled by 0.05, 400 updates (819k steps) for LunarLander (250 for the ring). Frozen policies are evaluated greedily (argmax) on 64 fresh episodes; the rings carry their persistent state into the evaluation.